High-Resolution Remote Sensing Image Scene Classification via Key Filter Bank Based on Convolutional Neural Network

Fengpeng Li, Ruyi Feng, *Member, IEEE*, Wei Han, and Lizhe Wang, *Senior Member, IEEE*

***Abstract*— High-resolution remote sensing (HRRS) image scene classification has attracted an enormous amount of attention due to its wide application in a range of tasks. Due to the rapid development of deep learning (DL), models based on convolutional neural network (CNN) have made competitive achievements on HRRS image scene classification because of the excellent representation capacity of DL. The scene labels of HRRS images extremely depend on the combination of global information and information from key regions or locations. However, most existing models based on CNN tend only to represent the global features of images or overstate local information capturing from key regions or locations, which may confuse different categories. To address this issue, a key region or location capturing method called key filter bank (KFB) is proposed in this article, and KFB can retain global information at the same time. This method can combine with different CNN models to improve the performance of HRRS imagery scene classification. Moreover, for the convenience of practical tasks, an end-to-end model called KFBNet where KFB combined with DenseNet-121 is proposed to compare the performance with existing models. This model is evaluated on public benchmark data sets, and the proposed model makes better performance on benchmarks than the state-of-the-art methods.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), key filter bank (KFB), scene classification.**

I. Introduction

High-resolution remote sensing (HRRS) image scene classification which aims at attaching semantic categories to HRRS images automatically is one of the fundamental tasks for Earth observation. Due to the rapid advancement of the technology of remote sensing [1], [2], a huge amount of HRRS images is acquired every day [3], [4], and HRRS images have detailed structure and abundant spatial patterns. Because of these advantages, HRRS image scene classification has been widely used in change detection [5], [6], urban planning [7], [8], and landcover/land-use classification [1], [9]. Due to the complex composition and variable scale, as shown in the second row of Fig. 1, HRRS image scene classification is a challenging work [10].

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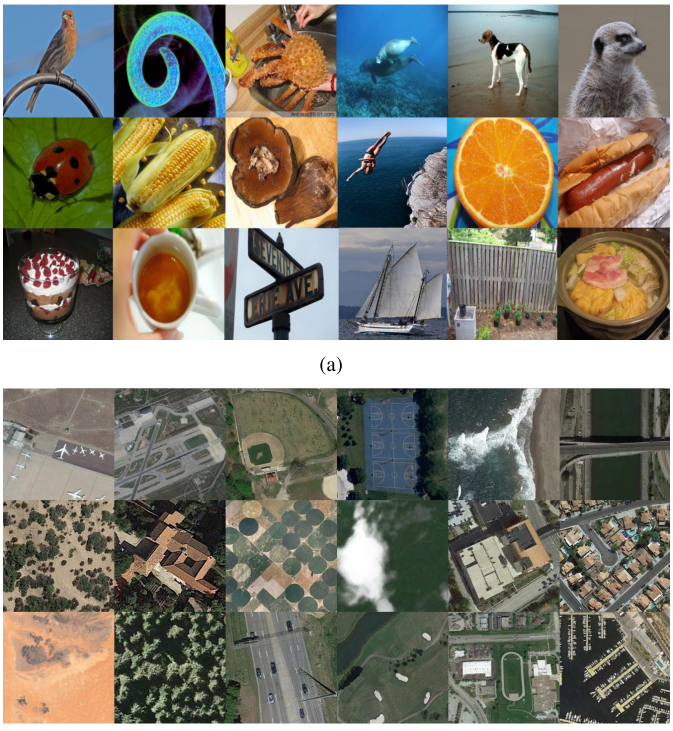

(a)

(b)

Fig. 1. Comparison between natural images and HRRS images. (a) Images selected from natural images scene data set ImageNet [11]. (b) Images selected from NWPU-RESISC45 [12].

Over the past decades, HRRS image scene classification has made great progress. In the early time, methods for HRRS scene classification were directly transformed from approaches for natural images, which led to terrible performance because of the significant gap between natural images and HRRS images, as shown in Fig. 1. To face this challenge, a large number of models based on the characteristics of HRRS images for scene classification have been proposed in the past several years [4], [13]–[17]. Nevertheless, an efficient and accurate model for HRRS image scene classification is still a difficult task [18]. In recent years, artificial intelligence, especially deep learning (DL) [19], has achieved rapid development. The typical DL models based on convolutional neural network (CNN), for instance, AlexNet [20], VGGNet [21], GoogleNet [22], and ResNet [23], have been widely applied in computer vision (CV) tasks, such as scene clssificaiton

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Fig. 2. Only the regions with dense vehicles are helpful for classification and the other regions are redundant.

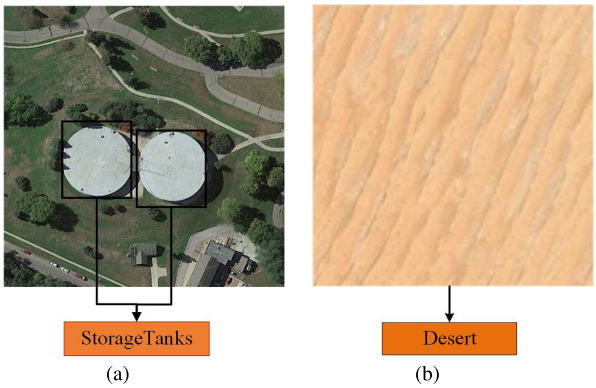

Fig. 3. Scene classification of HRRS images belonging to different categories depending on different features. (a) Tank determined by location information. (b) Desert determined by global features.

[24]–[26] on ImageNet [11] and made satisfying performance. Inspired by these achievements, much research work based on CNN [10], [12], [27]–[33] has been studied on HRRS image scene classification in the past several decades. Compared with traditional models for HRRS image scene classification, approaches based on CNN can extract more class-specific discriminative features to improve the accuracy of HRRS image scene classification.

Although these models based on CNN have achieved great performance on benchmark data sets, there are still some issues in current models. HRRS images are generated by a unique capturing mode, which is quite different from natural images. HRRS images always contain multiple categories of ground objects, and all spatial information is not equally useful. As shown in Fig. 2, this HRRS image is a scene of a parking lot. The regions with dense vehicles determine the category of the image, and regions of trees, roads, buildings, and bareland in the image are redundant. Most of the existing models capture the global features and treat these features with an equal contribution [12], [15], [34]–[37]; this interferes with class-specific discriminative features and leads to improper classification. In contrast, overemphasizing information [10] from key regions or locations can also ignore the importance

Fig. 4. Attention mechanism for recognizing the category of an HRRS image. (a) For regions with airplanes, it is essential for determination. (b) Key regions that are much brighter than other regions can be caught by our eyes because of the attention mechanism.

of global features. For example, based on the key regions in Fig. 3(a), it can be categorized as a scene of the tank, while the label of image (b) in Fig. 3 is determined by global features. Yuan *et al.* [38] proposed a method based on CNN to rearrange local features and global information to improve the performance of CNN models on HRRS image scene classification. The performance of rearranged policy for local features is further improved by ensemble CNN models [39], [40], which use several pretrained CNN models as feature extractors and combine the features to classify the scene of HRRS images. However, ensemble CNN models can be time-consuming in some cases. The reason is that there are a lot of hyper-parameters are needed to be set by a lot of experiments at the expense of a huge amount of time.

The root leading to the disadvantages of the above-introduced approaches is that these models do not correctly handle the relationship between global information and local information. As is known, when humans try to attach a category to an image, it is common for humans to select important regions or locations from an image and retain global sematic information at the same time. This is an essential property of the human visual system (HVS) called attention mechanism [41]. That is quite an intelligent procedure, and it can drive humans to get rid of unnecessary information and recognize objects efficiently. The image in Fig. 4(a) is a scene of the airport; the regions with airplanes have greater weights than other regions which are much brighter in Fig. 4(b); and Fig. 4(b) adds resemble focus of the range of human view. To imitate the mechanism to achieve a high-precision classification or detection, some models based on CNN combined with attention mechanisms are proposed. These models have a powerful representation ability and can capture key regions or locations. However, all these existing models are not able to retain global features, which causes loss of information. And the structure of attention mechanism is often complicated.

Facing the abovementioned challenges and avoiding adopting an overly complex structure, this article proposes a filter bank called key filter bank (KFB) without an attention mechanism structure (as shown in Fig. 5) to capture the features from key regions or locations and retain global information in this proposed method. Different pretrained CNNs are employed as feature extractors to obtain a high-level representation

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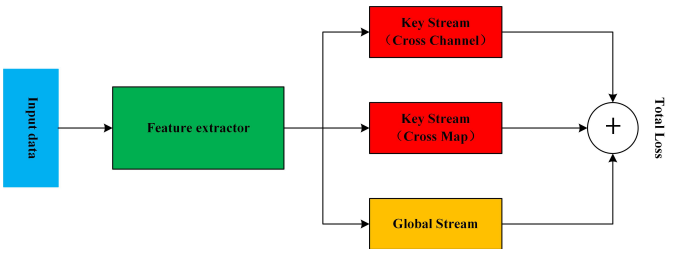

Fig. 5. Brief structure of KFBNet.

of HRRS images, and then the features are handled by KFB. The KFB captures the most class-specific discriminative information from the features and retains global information to make a predicted label at the same time. An end-to-end model combining CNN pretrained feature extractors and KFB, namely, KFBNet, is designed, which means it can be trained directly without any other operation. By the aid of this combined model, the hyper-parameters do not need to be used. Thus, the proposed model can effectively and efficiently achieve better performance.

The main contributions of this article can be summarized as follows.

1) A powerful asymmetric filter bank called KFB is proposed. The filter bank can capture class-specific features and retain global information at the same time.
2) A model combined KFB with DenseNet-121 [42], namely, KFBNet, which is trained by the end-to-end approach is proposed and applied in HRRS image scene classification. It improves the performance of CNN on HRRS scene classification task.
3) The proposed model uses a single CNN for feature extraction and no hyper-parameters are involved. Compared with the traditional CNN-based models without any changes, KFBNet makes a significant improvement.

The rest of this article is organized as follows. Section II presents related works. Section III describes the proposed method in detail. Experiments and analysis of benchmarks are conducted in Section IV. Section V concludes this article.

II. Related Works

A. HRRS Image Scene Classification

An enormous number of studies have been done on HRRS image scene classification because of its wide applications. There are mainly three kinds of scene classification models based on the feature extraction methods: handcrafted feature models, unsupervised feature models, and DL feature models which are as follows.

1) *Handcrafted Feature Models:* Handcrafted feature models are the earliest models applied in HRRS image scene classification. Histograms for color space and texture descriptors [43] are widely used in scene classification. Features extracted by these methods are global features, and they can be utilized to classify directly. However, they do not consider spatial information about original images, which probably leads to misclassification when images have similar global features extracted by these approaches, while the spatial information is quite different. For location of handcrafted feature models, scale-invariant feature (SIFT) [44], [45] and oriented gradients histogram [46], [47] are the extensively used methods which produce global representation by the mid-level descriptor. However, a single handcrafted feature tends to capture a category of specific features, which result in severe loss of information. To improve the performance of handcrafted feature methods, combining several kinds of handcrafted methods are considered [4], [17], [47]. For instance, Zhu *et al.* [4] proposed a method that can combine multiple features by fusion histogram level features to classify the bag-of-visual-words scene. Although these models based on handcrafted features exhibit good performance during image scene classification in early time, with the increasing of the requirement of image scene classification, the ability of handcrafted features cannot satisfy the demand due to the weak representation and overcomplicated combination mode.
2) *Unsupervised Features Models:* To solve the issues of handcrafted feature models, unsupervised feature learning methods are developed to make better performance on scene classification of remote sensing images [14], [18], [48]–[50]. Instead of designing feature descriptors like handcrafted feature learning methods, unsupervised models focus on learning feature encoding functions to represent categorizing data as learned features. These learned features are more conducive and discriminative to scene images classification compared with handcrafted features. K-means clustering [51], sparse coding [52], principal component analysis (PCA) [53], and auto-encoder are the typical unsupervised feature learning methods. Although these unsupervised feature learning approaches perform better on scene classification than handcrafted feature models, it is difficult for unsupervised feature learning methods to capture the most discriminative feature between different categories.
3) *DL Features Models:* Thanks to the rapid development of the capacity of high-performance computing and the available large-scale data set, DL methods that possess the powerful ability of feature acquisition have drawn a lot of attention [54], and many works based on DL make great achievements on several kinds of tasks, for example, object detection [55]–[57] and voice recognition [58], [59]. Compared with handcrafted features models, DL models can learn generalization features automatically just by feeding labeled inputs without complex engineering work for feature descriptors. And, features constructed by DL models based on CNN are more class-specific than unsupervised methods and handcrafted feature extractors, which promote the accuracy of HRRS image scene classification. However, DL models are quite difficult to obtain, and the requirement of data size and quality are extremely strict. To ease the difficulty and accelerate the training procedure, pretrained CNN models, which are already trained on other large-scale data sets like ImageNet, are transferred to other tasks [60]–[62].

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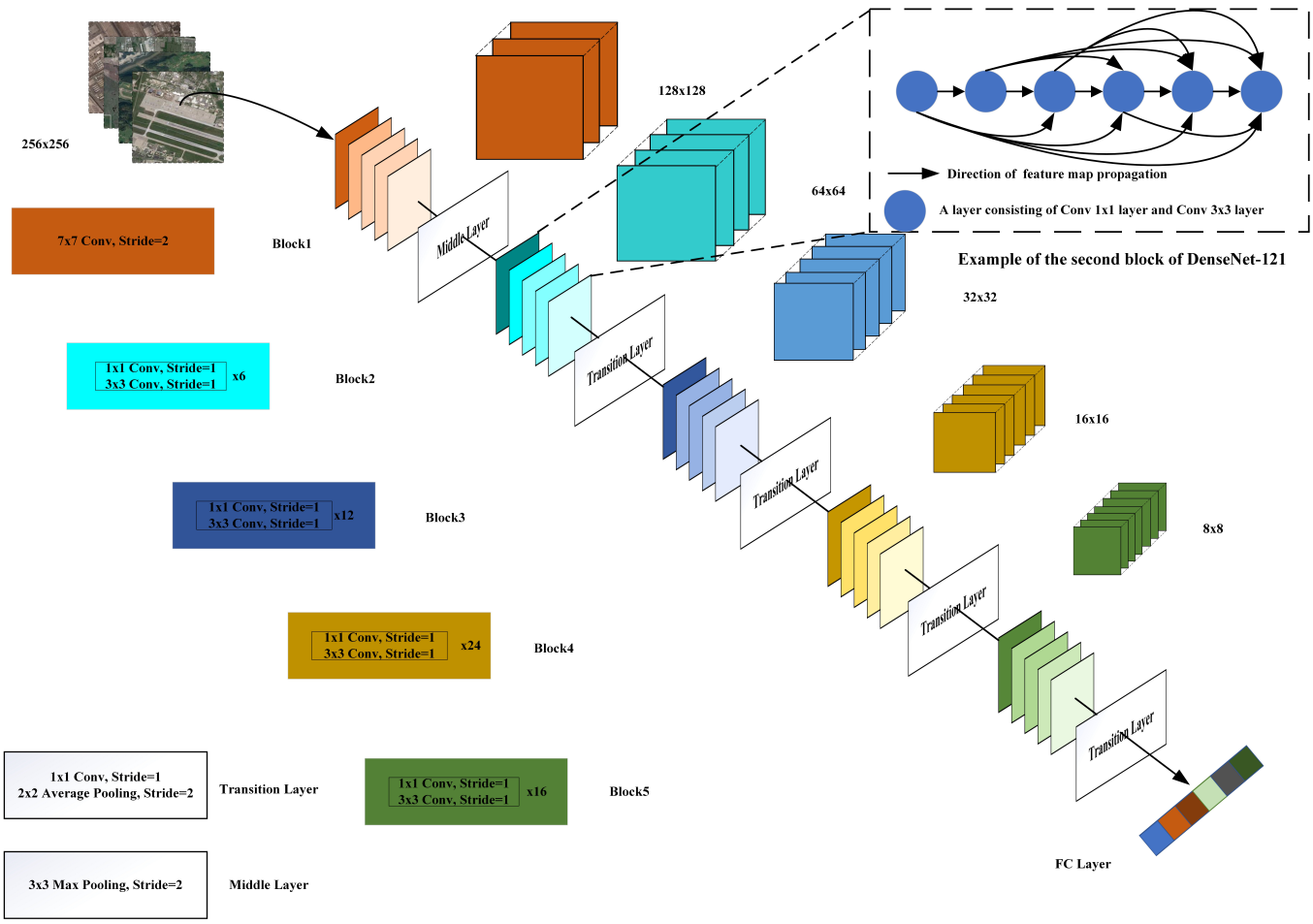

Fig. 6. This is a brief structure of the feature extraction part from DenseNet-121. There are five blocks in DenseNet-121. Transition layers concatenate preceding feature maps from each layer in a block; the arcs express the propagation direction of feature maps in a block; the numbers near the feature maps represent the width and height of feature maps; and the rectangle corresponding to each block shows the internal details of the block.

B. Attention Mechanism

Attention mechanism has already become an essential concept in DL, and it has been widely used in natural language processing (NLP) [63], [64] and CV tasks [65], [66]. The intuition behind the attention mechanism is human biological systems, and there are two kinds of attention mechanisms: soft attention mechanism and hard attention mechanism. For soft attention mechanism, there are flexible weight numbers, while hard attention mechanism has fixed weights with 0 and 1, which makes soft mechanism more adaptable to practical tasks.

There are a few existing models based on the attention mechanism for HRRS image scene classification. Wang *et al.* [10] proposed an HRRS image scene classification method based on recurrent attention combined CNN. This is the first successful attention model for HRRS image scene classification. Fang *et al.* [67] put forward a spectral-wise attention mechanism for hyperspectral images, which uses dense CNN to extract features. Despite these works, saliency detection is also famous for HRRS image scene classification [49], [68], [69] which aims at detecting salient regions or locations from HRRS images based on the assumption that regions of interest are always salient. Zhang *et al.* [49] proposed a sampling method based on the saliency-guided strategy to remove redundant features to extract the representation of HRRS images succinctly.

Although saliency detection methods can capture the features of salient regions or locations in an image, not all salient parts have equal significance for scene classification. Compared with saliency detection models, it is quite flexible for attention mechanisms to update its parameters by supervised training procedure to adapt the changes in the input to capture the regions of interest. The attention mechanism can achieve better performance during HRRS image scene classification if it can properly combine global information and local information.

III. Proposed Method: KFBNet

In this section, the details of the proposed methods are introduced, and each component of the KFB is explained. There are three parts to this section. The first part is an introduction to the feature extractors, where the reasons why CNNs are used as feature extractors for KFB are explained; the second one is the detailed description for the structure

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of KFB, while the last part is an explanation for KFBNet. The main idea of KFB is added as an extra part after feature extractor, which is used to capture the most discriminative class-specific features and retain global information at the same time.

A. Feature Extractor

The traditional feature extracted methods, such as histogram of oriented gradient (HOG) and local binary pattern (LBP), do not have excellent performance on HRRS image scene classification compared to DL feature extractors. To overcome the shortage of traditional feature extractors, DL feature extractors, especially CNN feature extractors, for example, AlexNet [20], VGGNet [21], GoogleNet [22], ResNet [23], and so on, have been widely used in feature representation of images currently due to their strong representation capacity [70]. The representation capacity and extracted features of different CNN feature extractors are significantly different. Compared with traditional feature extractors, features extracted by CNN are more discriminative, and compression of data is more reasonable. Hence, CNN features extractors are employed to combine with KFB in our work to achieve better performance on HRRS image scene classification. However, training a model without prior knowledge is difficult, especially when the amount of training data is limited. The pretrained policy can ease the problem of overfitting caused by a lack of training data and accelerate the convergence of models' parameters. Thus, the pretrained policy is adopted in our article. Our proposed KFB can combine with different feature extractors based on CNN. As far as we know, pretrained DenseNet is the single feature extractor that performs best on benchmark data set [40] without any change in the existing literature. KFBNet combined with DenseNet-121 is used as an example to introduce our proposed method.

DenseNet is a variant of ResNet, which can better solve the problem of gradient vanishing. Compared with ResNet, DenseNet strengthens the propagation of features, and make the best use of features from different levels in each block. At the same time, it can significantly reduce the number of parameters with the same number of layers compared with ResNet. For DenseNet-121 as displayed in Fig. 6, five blocks consist of several batch normal convolutional layers with activation ReLU for each block. Between two blocks, there is a transition layer that combines preceding feature maps from the same block which are used as input for the next block. In the last part of the model, there are fully connected layers with activation ReLU and dropout layers for classification. ReLU is an update activation function for traditional sigmoid, and it is a half-wave rectifier function, which can retain positive numbers and make the negative numbers to zeros. Compared with sigmoid, ReLU can relieve the vanishing gradient problem.

In the proposed method, a pretrained DenseNet-121 model is taken from the Pytorch model library, which has been trained on ImageNet. KFB uses the features after training from the four blocks of DenseNet-121. In this procedure, the input size is $3 \times h \times w$, and the size of the feature map is

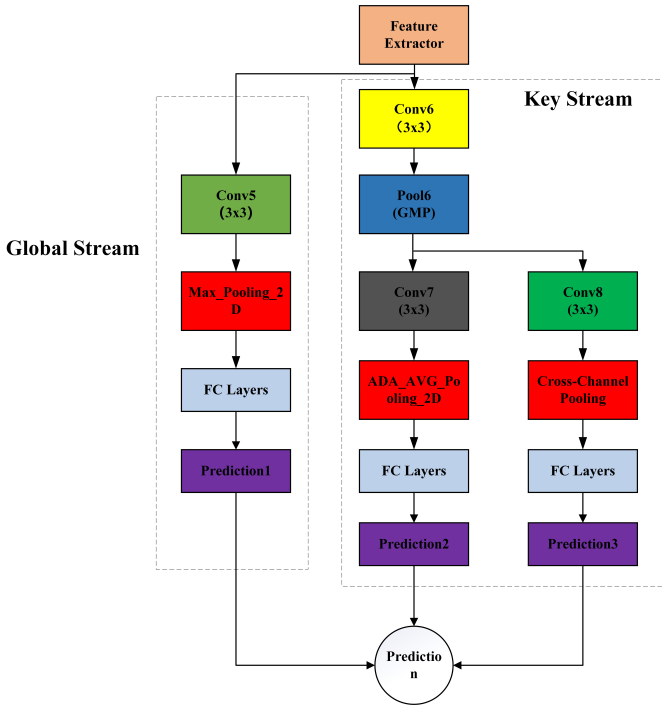

Fig. 7. KFB consists of two streams, three branches. The left branch is G-Stream. The middle branch and right branch are K-Stream.

$256 \times h/16 \times w/16$

$$F = (f_1, f_2, \ldots, f_i), \quad i \in [1, 1024]. \tag{1}$$

F is feature maps extracted by selected pretrained DenseNet-121 from the input HRRS images, and f_i is the ith feature map of F. All these feature maps will be processed in KFB to capture the key features of regions or locations and obtain global information correctly to predict scene labels for input HRRS images.

B. Structure of KFB

The structure of KFB, as shown in Fig. 7 is the core part of the proposed method in this article. As the previous introduction, most of the existing models for HRRS image scene classification tend to capture global features of the images and use fully connected layers to make predicted categories for the inputs. This is severely disturbed by redundant information from irrelevant regions or locations. Existing attention mechanisms for HRRS images cannot retain global information. Thus, if a structure can extract discriminative class-specific features, get rid of the unrelated information, and retain global information, it can significantly promote the performance of the CNN models on HRRS image scene classification.

To achieve this, KFB is supposed to be designed sophisticated. The left branch (displayed in Fig. 7) in the KFB is Global Stream (G-Stream), which is made up of a convolutional part and a fully connected block. The convolutional part is the last block from pretrained DenseNet-121. Note that, if the feature extractor is based on ResNet-50, the convolutional part is the last block of ResNet-50. VGGNet-16 is

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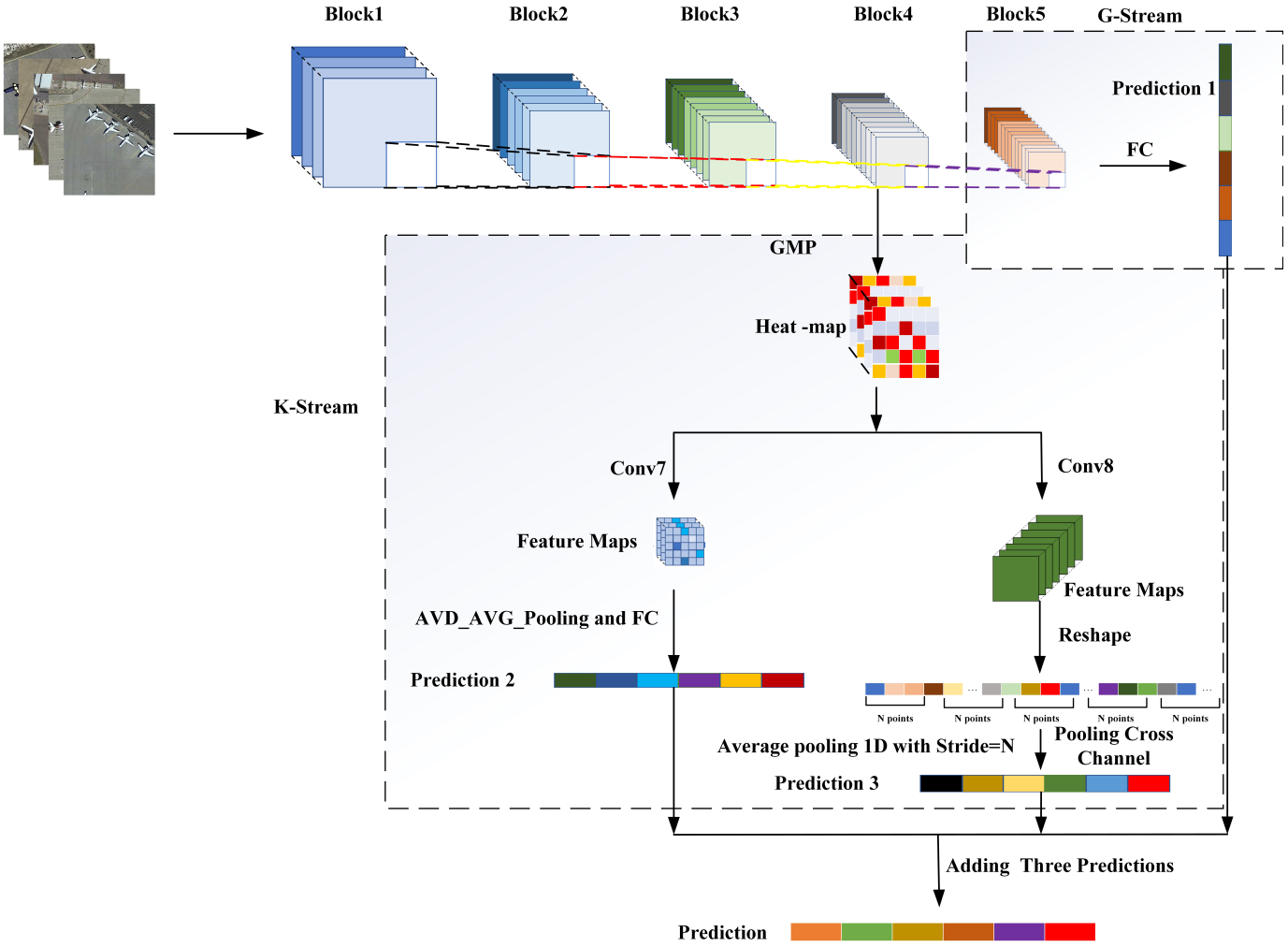

Fig. 8. Structure of KFBNet based on DenseNet-121 and the N in the picture is equal to k.

used in the same way. The feature maps from selected feature extractor without any operation on them contains global information. The G-Stream uses the feature maps to make predicted categories called prediction1 for input data by fully connected layers where there are three linear layers with ReLU and Dropout in the first two layers and without any other operation in the last linear layer. The G-Stream is quite essential for the classification of HRRS images, which significantly depend on global information, as shown in Fig. 3(b). For the HRRS images in Fig. 3 (a), capturing features from class-specific regions or locations and getting rid of redundant information are the points for correct classification.

As shown in Fig. 7, the middle and right branches are combined to capture the most discriminative features from feature maps. The parameters of each part in Key-Stream (K-Stream) are displayed in Table I. n_class is the number of categories for the training data. Conv6 is used to filter the feature maps where each pixel corresponds to a small region in the original images, and the values of the pixels with several channels are greater than ordinary pixels which represent key regions in original images. The Global Max Pooling (GMP) for feature maps is used to retain the most important features, in which the feature maps exported by GMP are called heat-maps. There are k channels features on the heat-maps for each category contained in the data set. According to our experimental analysis, k is set to 20 here. In this case, KFB performs well on each data set, and the time cost and graphics memory do not significantly change at the same time. The middle branch takes Conv7 to transform

TABLE I
PARAMETERS OF EACH PARTS IN K-STREAM

Part Name	Kernel	Stride	Channel	Input_size
Conv6	3×3	1×1	20×n_class	32×32
Global Max Pooling	16×16	16×16	20×n_class	32×32
Conv7	3×3	1×1	n_class	2×2
Adaptive-Average Pooling	2×2	2×2	n_class	2×2
Conv8	2×2	1×1	20×n_class	2×2
Cross-Channel Pooling (1D)	20	20		20×n_class

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the number of heat-maps' channels to the category of categories, and then the feature maps are imported to Adaptive-Average Pooling (ADA_AVG_Pooling_2D) which computes the average value for each channel of feature maps. Finally, the extracted features are reshaped to 2-D after the operation of Adaptive-Average Pooling called prediction2. However, the middle branch only handles the features with a single channel for each category, which may lead to misclassification if irrelevant regions or locations have the same response information with regions or locations containing class-specific features in a single channel.

To make each channel of the feature maps respond to the features from key regions and avoid confusion information from irrelevant regions, the right branch is utilized. The right branch uses the Conv8 with the BatchNorm layer and ReLU followed by 2×2 Max Pooling with stride $= 1$ to adjust feature maps, which measures $20 \times n_\text{class} \times 1 \times 1$ after the operation of Conv8. In the feature maps, there are 20 response channels for each category, and 1-D Cross-Channel Pooling is used to compute the average value of 20 channels for each category. The average response for each category is prediction3.

In each branch, Average Pooling is used instead of Max Pooling for each pooling procedure, as a balanced response for each category of categories is desired. Similarly, pooled filters based on Average Pooling can be affected during backpropagation, while pooled filters from Max Pooling only make differences in maximum response [71].

C. KFBNet

As shown in Fig. 8, KFBNet is a CNN model combined with DenseNet-121. The performance of KFBNet based on DenseNet-121 is compared with other models in benchmark data sets in Section IV. KFBNet uses the first four blocks of DenseNet-121 to extract the feature maps of input data. Each branch of KFB takes the feature maps to make predicted categories for inputs. During training, the predictions for each branch are regularized with practical labels by CrossEntropy loss function. The parameters of KFBNet are updated by using total loss through backpropagation, while the predicted categories are the sum of predictions for each branch in the test process.

In addition, the probability of softmax function, which is used to make prediction, is defined as

$$P_i^n = \frac{\exp(W_i)}{\sum_{j=1}^{L} \exp(W_j)}, \quad i \in 1, 2, \ldots, n \in 1, 2, 3 \tag{2}$$

where W_i is the weight for the ith label and n is the number of branches from left to right. L is the number of the total categories of labels. The prediction categories of nth branch are

$$V_n = (v_1, v_2, \ldots, v_L) \tag{3}$$

where V_n is the prediction vector. CrossEntropy is applied to calculate the distance between predicted label vector and true label vector, which is shown as follows:

$$\text{loss}_n = -\sum_{h=1} y_h^t \log(y_h), \quad h \in 1, 2, \ldots, H \tag{4}$$

where y_h^t is the number of true label vectors while y_h is hth predicted vector. The total loss used to update the parameters of KFBNet can be calculated by

$$\text{total_loss} = \sum_{j=1}^{n} \text{loss}_j, \quad j \in 1, 2, 3. \tag{5}$$

The prediction in test process is computed as

$$\text{prediction} = \sum_{j=1}^{n} \text{prediction}_j, \quad j \in 1, 2, 3. \tag{6}$$

IV. Experiment

In this section, different benchmark data sets will be utilized to evaluate the effectiveness of the proposed method with existing methods. First, benchmark data sets and evaluation metrics are to be introduced. Second, the details of experiments, for instance, parameters setting and selection for feature extractor, are analyzed. In Section IV-D, comparisons of performance on benchmarks with state-of-the-art methods are displayed to show the promotion of the proposed method in this article.

A. Experimental Data Set

1) UC Merced (UCM) Land-Use Data Set: UCM Land-Use data [72] is a widely used benchmark data to test the effectiveness of scene classification approaches, and it is the first ground truth data set generated from available HRRS images. All these images are extracted from aerial orthography. There are 21 typical land-use categories, including dense residential, freeway, harbor, airplane, baseball diamond, medium-density residual, forest, chaparral, beach, sparse residential, tennis courts, storage tanks, river, runway, mobile home park, overpass parking lot, intersection, golf course, agricultural, and building. For each category, there are 100 images measuring 256×256 with a 30-cm spatial resolution containing three RGB channels. It is a challenging task to classify different categories on UCM due to the limited numbers of images in each category, and there are some categories that are quite similar to each other, like medium and residential and dense residential areas, which easily results in misclassification. To compare with the existing methods [10], [29], [32], [35], [73], 80% of the images in each category for training are used to train, and the rest of the images are used to test.

2) Aerial Image Data Set (AID): AID [36] is a large-scale land-use data set for HRRS images consisting of 30 scene types: airport, baseball field, bare land, bridge, beach, center, commercial, church, dense residential, farmland, desert, industrial, forest, meadow, mountain, medium residential, parking, playground, park, pond, railway station, port, river, resort, sparse residential school, square storage tanks, viaduct, stadium, and foot field. There are 10 000 images with 600×600pixels for each image. All these images are labeled by remote sensing image interpretation specialists. We perform two group experiments with 20% and 50% data for training and the rest are used as test data, respectively, according to the existing models in [10], [32], [39], [74], and [75].

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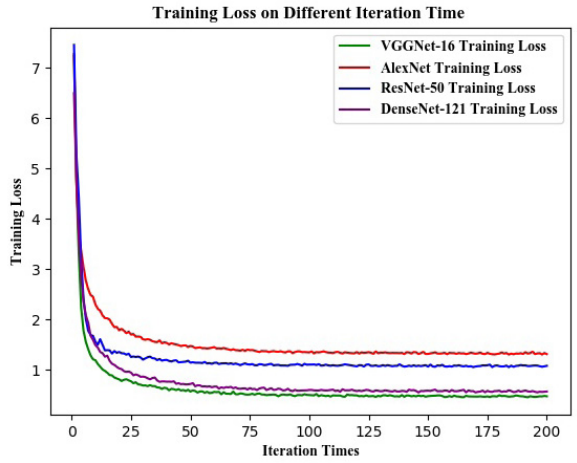

Fig. 9. Comparison of training loss for KFBNet with different feature extractors in each training iteration.

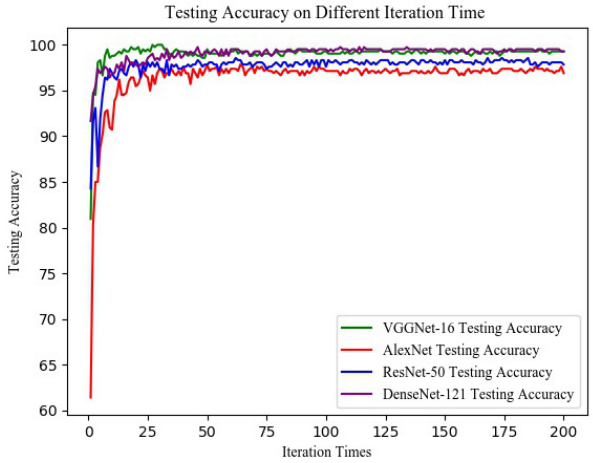

Fig. 10. Comparison of performance for KFBNet with different feature extractors in each training iteration.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT k VALUE ON UCM LAND-USE DATA SET

k Value/Ratio	OA(80%)
1	99.41±0.12
10	99.31±0.23
20	99.88±0.12
30	99.88±0.20

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT VARIATION FREQUENCY OF LEARNING RATE VALUE ON UCM LAND-USE DATA SET

lr VARIATION FREQUENCY/Ratio	OA(80%)
1	99.19±0.12
2	99.88±0.12
3	99.53±0.01
4	99.31±0.23
Fixed Learning Rate	99.48±0.11

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT FEATURE EXTRACTORS ON UCM LAND-USE DATA SET

Models/Ratio	OA(80%)
Pre-trained AlextNet with SVM	94.58±0.11
VGG-VD-16 [10]	97.26±0.60
KFB(AlextNet)	97.26±0.60
KFB(VGGNet-16)	99.76±0.24
KFB(ResNet-50)	98.92±0.18
KFB(DenseNet-121)	**99.88±0.12**

TABLE V
PERFORMANCE COMPARISON OF MODELS ON UCM LAND-USE DATA SET

Models/Ratio	OA(80%)
Pre-trained VGGNet-16+SVM	97.14±0.10
Combing Scenatios I and II [35]	98.49
D-CNN [32]	98.93±0.10
Fusion by Addition [29]	97.42±0.17
CNN-NN [73]	97.19
ARCNet-VGG16 [10]	99.12±0.40
KFBNet(VGGNet-16)	99.76±0.24
KFBNet(DenseNet-121)	**99.88±0.12**

3) OPTIMAL-31 Data Set: OPTIMAL-31 data set [10] is a new data set for scene classification, and all the images in this data set are collected from Google Earth. There are 31 categories and 1860 images in this data set. And, there are 60 images with 256 × 256 for each category, and the categories in these data are airplane, baseball field, airport, beach, basketball court, bushes, bridge, bushes, freeway, church, business district, round farmland, desert, forest, playground, factory, meadow, golf field, harbor, crossroads, lake, island, medium houses, mountain, mobile house area, overpass, square farmland, runway, parking lot, roundabout, and runway. The setting rule of OPTIMAL-31 is more reasonable than other data sets, and the categories of scenes are more abundant. To compare with existing approaches that have displayed the effectiveness on OPTIMAL-31, we use

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TABLE VI
PERFORMANCE OF EACH BRANCH OF KFBNET ON UCM LAND-USE DATA SET

lr VARIATION FREQUENCY/Ratio	OA(80%)
Left branch	98.02±0.17
Middle branch	99.45±0.40
Right branch	99.02±0.45
Left+Middle	99.33±0.19
Left+Right	99.35±0.31
Middle+Right	98.52±0.05
Letf+Middle+Right	99.88±0.12

80% of the total data set to train the model, and the rest is used in the test procedure, according to [10].

4) NWPU-RESISC45 (NWPU-45) Data Set : NWPU-RESISC45 [12] is a challenging data set for HRRS scene classification. There are 45 species of HRRS images with a size of 256 × 256 in RGB color space, and each category contains 700 images, and there are a total of 31 500 images in the data set. The spatial resolution of this data set varies from 0.3 m to 0.2 m/pixel, and the 45 categories include airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. As far as we know, this is the largest data set for HRRS image scene classification [40]. There are abundant categories in this data set and abundant as well as variable images in each category. Images belonging to different categories are quite similar to each other. All the properties of this data set make it a challenge to achieve satisfying performance. According to [30], [32], [33], [39], [40], and [76], there are two groups of experiments for NWPU-45 with 10% and 20% images for training, and the models are tested on the rest of the data.

B. Evaluation Metric

For image scene classification, there are three widely used evaluation metrics, including confusion matrix (CM) [12], overall accuracy (OA), and average accuracy (AA), which are as follows.

1) *Confusion Matrix:* CM is widely used in supervised classification to evaluate the performance of models on each category in a visual way. In the CM, the row represents the true labels, while the column illustrates the prediction categories for the inputs. Thus, it is easy to be observed from the CM whether the categories of test data are confused or not.

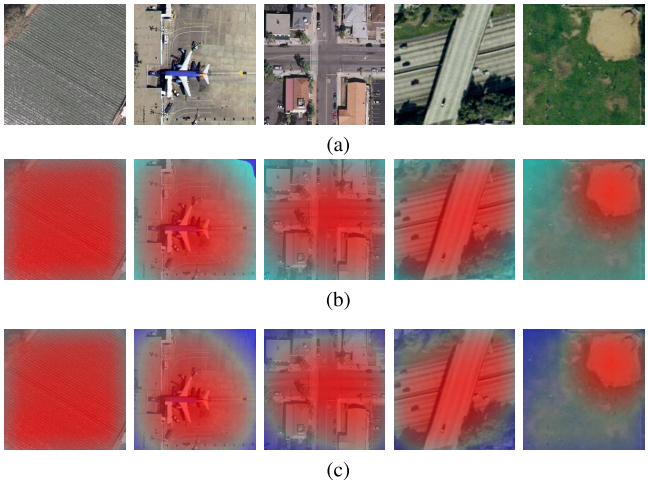

Fig. 11. Class-specific regions located by two streams of KFBNet. The depth of red mask color reflects the importance extent of different regions in HRRS images. (a) Original HRRS images in the testing data set. (Left to right) These images are categorized as agricultural, airplane, intersection, overpass, and baseball diamond. (b) Class-specific regions located by G-Stream. (c) Class-specific regions located by K-Stream.

TABLE VII
PERFORMANCE COMPARISON OF MODELS ON AID DATA SET

Models/Ratio	OA(20%)	OA(50%)
ARCNet-VGG16 [10]	88.75±0.40	93.10±0.55
Pre-trained VGGNet-16+SVM	89.33±0.23	96.04±0.13
Binary Patterns encoded CNNs [74]		95.73±0.16
Dense connectivity based two-steam fusion [75]		95.99±0.35
D-CNN [32]	90.82±0.16	96.89±0.10
HW-CNN [39]		96.98±0.33
KFBNet(VGGNet-16)	**94.27±0.02**	**97.19±0.07**
KFBNet(DenseNet-121)	**95.50±0.27**	**97.40±0.10**

2) *Overall Accuracy:* OA is computed as the numbers of test data that are corrected predicted divides the total numbers of images in each category.
3) *Average Accuracy:* AA is the average value of accuracy for each category in test data sets.

Since all experimental data sets in this article have balanced numbers for each category, AA is equal to OA under and, thus, OA and CM are adopted to evaluate the performance of different models.

It is universally acknowledged that the number of training data significantly influences the performance of the model on the test data, and even different training data with the same ratios may lead to different achievements on different test data in the same data set. Therefore, we perform several experiments on each data set with different partition of training data set and testing data set to calculate the average accuracies to avoid one-time occasionally. We also use horizontal flipping, vertical flipping, and random rotation from −180° to 180° to augment training data.

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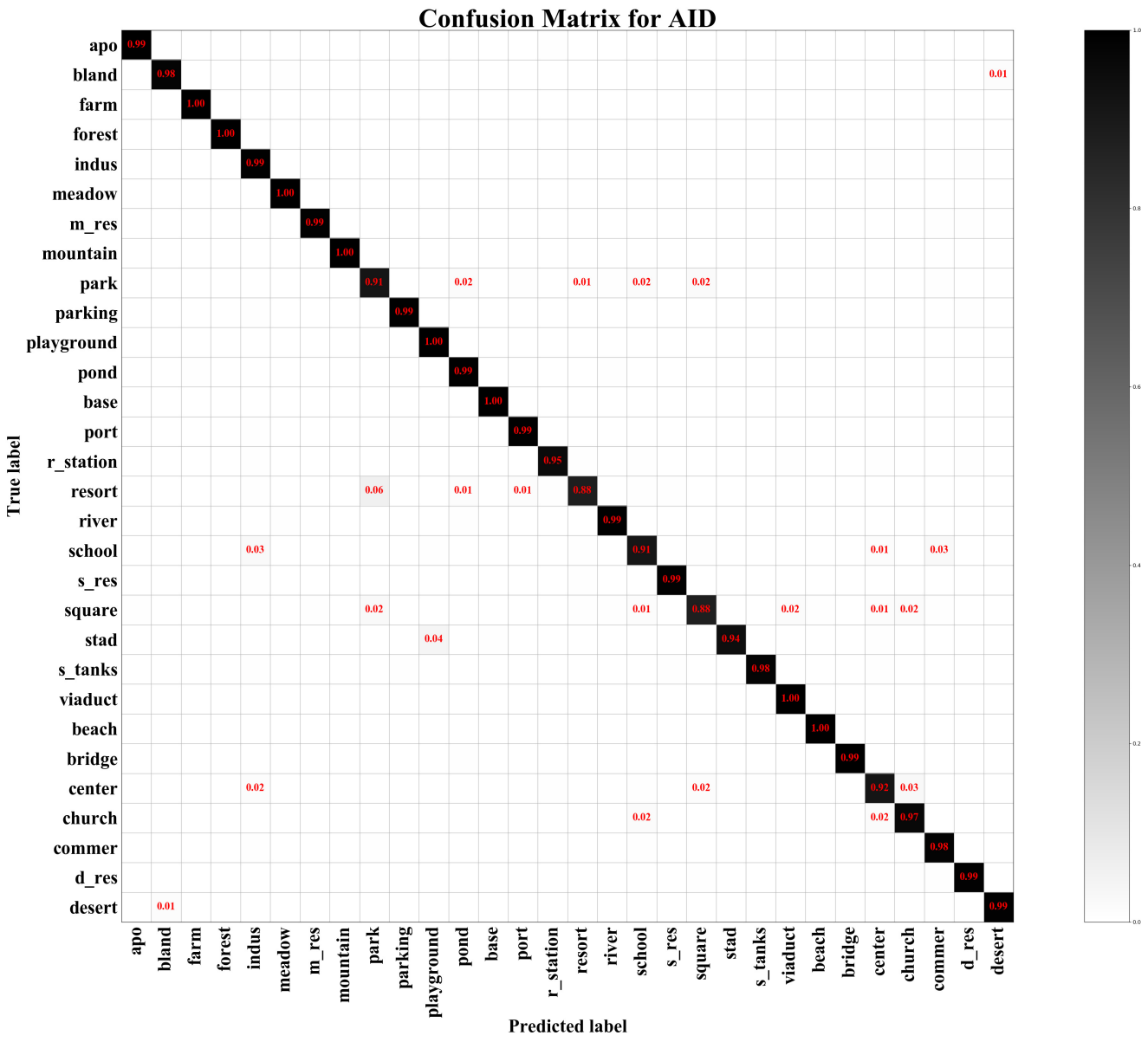

Fig. 12. CM of the proposed method KFBNet based on DenseNet-121 over AID data set.

C. Training Details

In this section, we introduce training parameters and also discuss the feature extractors used in experiments. In addition, the detailed setting of the experiments will be explained in the training and evaluation procedure which are as follows.

1) *Training Parameters:* To evaluate the proposed model, we use pretrained CNN models to extract the feature maps from HRRS images, whereas the pretrained weights of these models come from the model library of Pytorch. We also undertake a Pytorch implementation code to carry out experiments on a personal computer with i7-7700 CPU, 24-GB RAM, and NVIDIA Titan V. In the procedure, variable learning rate is used. The original learning rate lr is 1×10^{-4}, and if the testing accuracy on training procedure does not improve in one training step, lr is changed as $\text{lr} = \text{lr} * 0.9$, and the low limit of lr is 1×10^{-6}. The batch size is set to 18 for images with a size of 256×256 in 200 epochs. Moreover, to achieve the most remarkable performance over each data set, we use several experiments over the UCM data set to find the most appropriate k value.
2) *Convolutional Feature Extractor:* Different CNN models can generate different representations for the same images. The proposed method KFB can be combined with different CNN feature extractors. To find a suitable extractor with KFB, we test different feature extractors, such as VGGNet-16, AlexNet, ResNet-50, and DenseNet-121, with KFB on UCM Land-Use data set.

D. Performance

In this section, we display the performance of KFBNet on several benchmark data sets. Furthermore, the effect of

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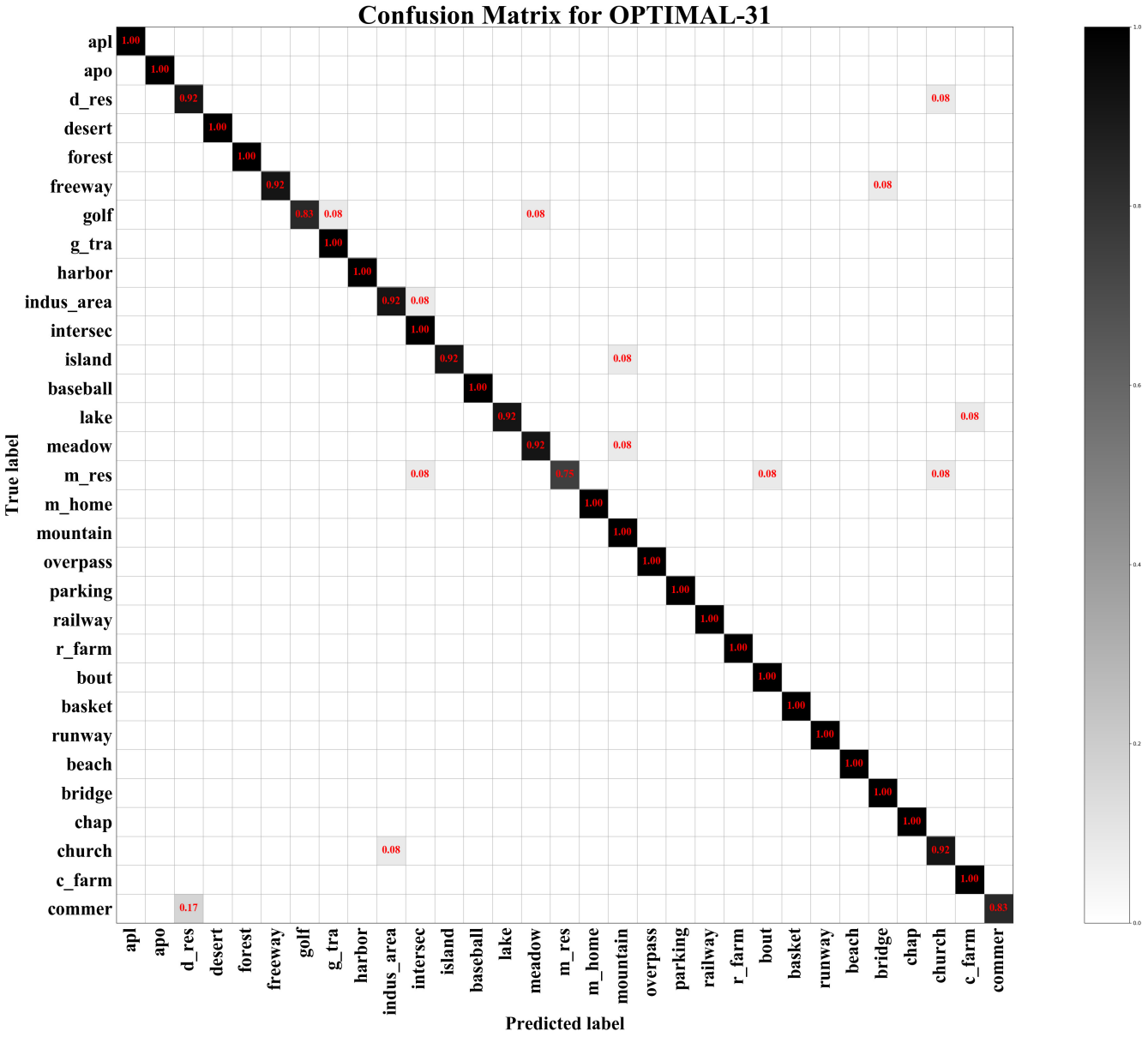

Fig. 13. CM of the proposed method KFBNet based on DenseNet-121 over OPTIMAL-31 data set.

different feature extractors is shown by combining KFB with different CNN models on the UCM data set.

1) UCM Land-Use Data Set: For the UCM data set, we test the KFBNet, which is combined with pretrained AlexNet, pretrained VGGNet-16, pretrained ResNet-50, and pretrained DenseNet-121. About 80% of the data set is used to train the model, and the rest is used to test the performance of KFBNet with different feature extractors. To find the most appropriate parameter setting of k, we use several experiments by KFB combined with DenseNet-121 over UCM with different k values. As shown in Table II, when $k = 20$, KFB with DenseNet-121 achieves the most excellent performance over UCM. Furthermore, based on [40], an updated learning rate lr based on certain frequency has a positive impact on the performance of the model, and we use experiments over the UCM data set to find the best parameter setting of lr. As shown in Table III, KFBNet with DenseNet-121 achieves the best performance when the updated frequency of lr is set to 2. Based on experiments over the UCM data set, we use $k = 20$ for KFBNet, and updated frequency is set to 2 for learning rate lr.

As displayed in Fig. 9, it can be noticed that the KFBNet combined with pretrained VGGNet-16 and with pretrained DenseNet-121 is more effective compared with ResNet-50 and AlexNet. From Fig. 10, we find KFBNet with pretrained VGGNet-16 has achieved the best performance on test data in 40 iteration times, and then the accuracy is around 99.52% and tends to be slightly lower than DenseNet-121. The results indicate that DenseNet-121 has better representation ability and much more stable than other CNN feature extractors.

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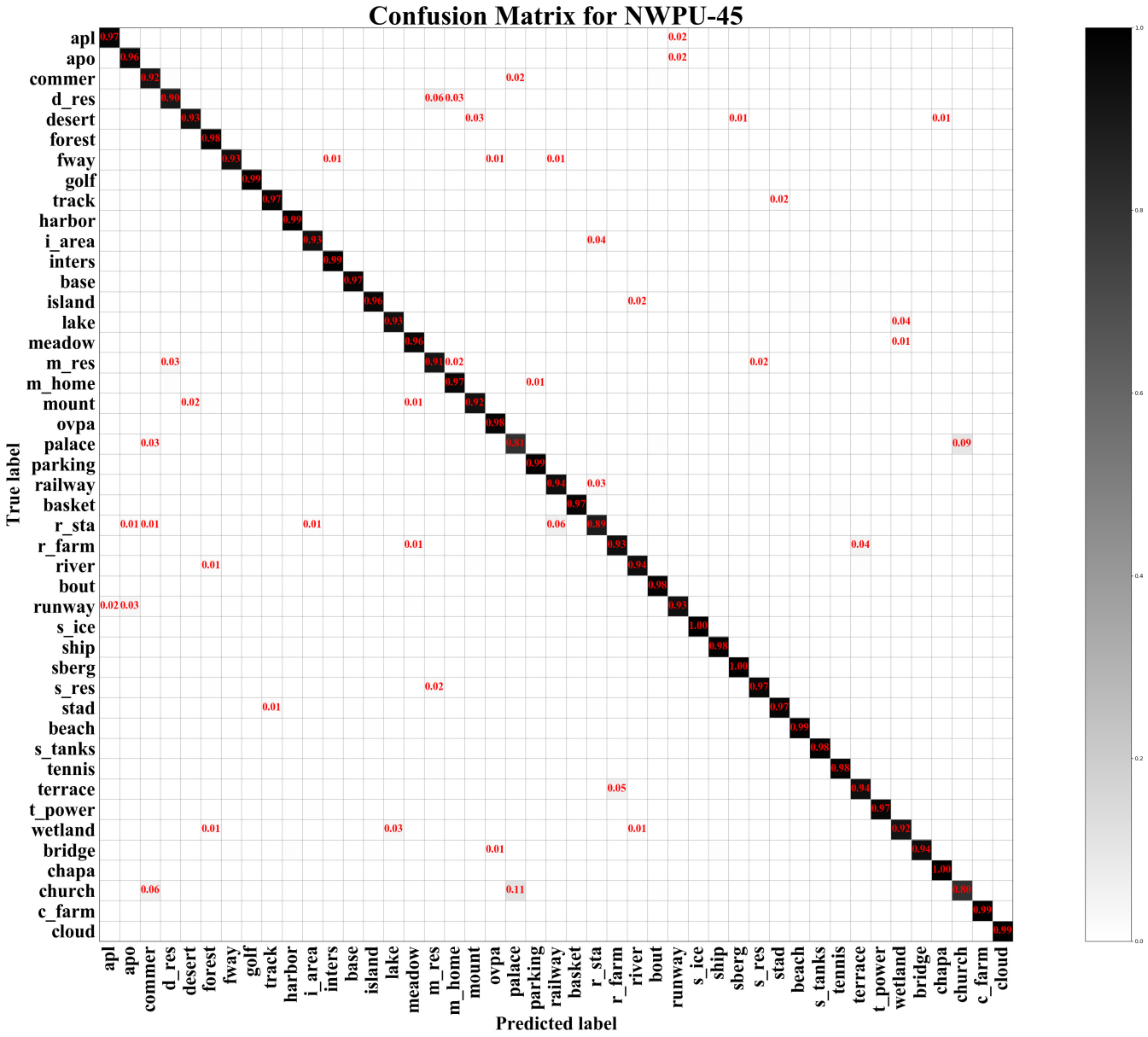

Fig. 14. CM of the proposed method KFBNet based on DenseNet-121 over NWPU-45 data set.

However, VGGNet-16 is the most efficient feature extractors, and it achieves the best performance in 25 training steps on the UCM data set.

Although the KFBNet combined with AlexNet or VGGNet-16 does not perform better than DenseNet-121 (as shown in Table IV), they still make better achievement than original CNN models, which indicates that KFB is sufficient for HRRS scene classification. As displayed in Table V, KFBNet based on DenseNet-121 makes state-of-the-art performance on scene classification on the UCM data set with 80% data for training. And the KFBNet with VGGNet-16 is still better than other previous methods. The accuracy of classification performed by KFBNet with DenseNet-121 and VGGNet-16 can be up to 100%. To better demonstrate the effectiveness of KFB, we take DenseNet-121 and VGGNet-16 as feature extractors and compare the performance of KFBNet with DenseNet-121 and KFBNet with VGGNet-16 with existing approaches on other data sets, as shown in Table V.

To analyze the function of each branch of KFBNet combined with DenseNet-121, we present performance of each branch and combinations of different two branches to display the effectiveness of KFBNet, shown as Table VI, and we also use visual heat maps generated by G-stream and K-stream used Grad-CAM [77] to show the class-specific feature capturing capacity of the two streams of KFBNet. As shown in Table VI, the performance of the combination of three branches is better than the prediction of each branch and combinations of two branches. Furthermore, visual heat maps, as shown in Fig. 11, displays that the position of class-specific regions located by K-Stream is much more accurate than G-Stream. And it

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TABLE VIII
PERFORMANCE COMPARISON OF MODELS ON OPTIMAL-31

Models/Ratio	OA(80%)
Fine-tuning AlexNet [10]	82.57±0.12
Fine-tuning GoogleNet [10]	87.15±0.45
Fine-tuning VGGNet16 [10]	87.45±0.45
VGG-VD-16 [36]	89.12±0.35
ARCNet-VGGNet16 [10]	92.70±0.35
KFBNet(VGGNet-16)	**95.12±0.13**
KFBNet(DenseNet-121)	**95.60±0.63**

TABLE IX
PERFORMANCE COMPARISON OF MODELS ON NWPU-45

Models/Ratio	OA(10%)	OA(20%)
Pre-trained VGGNet-16+SVM	87.15±0.45	90.36±0.18
IOR4-VGG16 [33]	87.83±0.16	91.30±0.17
SSGA-E [30]	88.60±0.95	
D-CNN [32]	89.22±0.50	91.89±0.22
Triplet networks [76]		92.33±0.20
Deeper ConvNet+Context Aggregation [39]		93.47±0.26
DenseNet-121 [40]	91.06±0.61	93.33±0.55
Hygra: an ensemble of CNNs [40]	92.44±0.34	94.51±0.21
HW-CNN [39]		94.38±0.16
KFBNet(VGGNet-16)	90.27±0.02	92.54±0.03
KFBNet(DenseNet-121)	**93.08±0.14**	**95.11±0.10**

means that G-stream introduces redundant information, which probably leads to misclassification.

2) Aerial Image Data Set: AID data set is much more complex than UCM; we test the performance of the KFBNet with DenseNet-121 and as well as with VGGNet-16 using 20% and 50% data for training, respectively. From Table VII, on two kinds of split ratios, KFBNet has achieved the best performance with DenseNet-121 acted as feature extractor. As shown in Fig. 12, KFBNet with VGGNet-16 is better than other previous models on AID, and KFBNet with DenseNet-121 obtains 100% classification accuracy for 8 out of 30 categories. With 20% data for training, KFBNet based on DenseNet-121 makes 4.68% improvement of accuracy compared with D-CNN and obtains 0.47% higher than HW-CNN under 50% data for training. The experimental result shows that KFB is effective in AID image scene classification, although this data set is complex, and the categories are quite similar.

3) OPTIMAL-31: There are 31 categories in the data set. Categories in the data set are more discriminative than other data sets. However, there are only 60 images for each category in OPTIMAL-31, which is still a challenging work to better classify them. We use 80% of data to train the proposed model, and the rest is used to evaluate the performance. The accuracy of testing experiments is quite unstable, as there are only 12 images for testing each category, which means that accidental factors may greatly affect the performance of KFB. From Table VIII, KFBNet with DenseNet-121 has got the highest accuracy 95.60% ± 0.63% on OPTIMAL-31 compared with other methods under the same ratios for training. And KFBNet with VGGNet-16 is still better than existing models. Fig. 13 displays the classification result for each category. From the experimental results, we find that KFB can deal with the situation that there are a few samples for training and testing, although the data set is complicated, and images belonging to same category might be easily confused. Furthermore, we find that there is a large fluctuation range of performance of KFB with DenseNet-121 over OPTIMAL-31, which is because of the limited number of samples in this data set. Due to the presence of only 12 images for each category in this data set for testing, it means that changes of classification results for a few samples in the testing data set will lead to a large fluctuation of OA.

4) NWPU-RSISC-45: We evaluate the model with 20% and 10% for training, and the rest is used to do testing. As shown in Table IX, KFB with DenseNet-121 makes the state-of-the-art performance on NWPU-45 with training ratios 10% and 20%. Compared with the state-of-the-art method, and Hydra on NWPU-45, KFBNet with DenseNet-121 achieves 0.64% improvement with 10% ratios for training and 0.60% improvement under 20% ratios of NWPU-45 for training. Fig. 14 exhibits the detailed classification results for NWPU-45 with 20% for training. Although KFBNet with DenseNet-121 achieves state-of-the-art performance, KFBNet with VGGNet-16 is inferior in quantitative results, due to the over-complication of NWPU-45 and poor representation ability of VGGNet-16. As shown in Table IX, pretrained DenseNet-121 has a better performance than pretrained VGGNet-16+SVM over NWPU-45, which indicates that DenseNet possesses a stronger capacity of representation. From Fig. 14, it can be found that images labeled as church and palace are easy to be confused, though KFBNet with DenseNet-121 performs excellently in general.

V. CONCLUSION

In this article, a novel filter bank, denoted as KFB, is proposed to improve the performance of image scene classification combined with a single CNN model. To better verify the effectiveness of the proposed method, an end-to-end model, namely, KFBNet, is developed, which utilizes DenseNet-121 as the feature extractor. Compared with state-of-the-art models, there are few hyper-parameters in the proposed KFBNet. Different from the traditional DL-based image scene classification network, there are two streams in the novel network: G-Stream and K-Stream, in which the G-Stream is used to predicted categories of input data for global information,

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and the middle branch of K-Stream can predict the label of input data cross the map of heat-maps, while the right branch of KFB is to forecast labels of input data by the average value of each channel for each category in the heat-maps. In addition, the KFB can be combined with different kind of CNN feature extractors, which indicates the extensibility of the proposed method.

To better evaluate the effectiveness of the proposed method, we use feature extractors based on different CNN models for image scene classification with different benchmark data sets and confirm that the pretrained DenseNet-121 combined KFB outperforms others. Still, compared with traditional models, KFB can improve the performance of CNN feature extractors significantly, which uses the same CNN feature extractors with KFBNet. However, for similar images that belong to different categories, KFBNet does not make a great difference. In the future, more studies will be made to solve the challenging problem that similar images are easy to be confused although they belong to different labels.

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